

Multiple Choice (10 marks)

1. The ROM programmed during manufacturing process itself is called
 - (a) PROM
 - (b) MROM
 - (c) DRAM
 - (d) EPROM
 - (e) SRAM
2. An electromagnetic wave of 40 KHz is radiated by an antenna . Find the time taken by this wave to reach a point 60 km away.
 - (a) 0.3 msec
 - (b) 0.8 msec
 - (c) 0.6 msec
 - (d) 0.25 msec
 - (e) 0.1 msec
3. Evaluate the Laplace inverse of $F(s) = [(5s^2 - 15s + 7) / \{(s + 1)(s - 2)^3\}]$
 - (a) $-e^{-t} - e^2 t + 2te^2 t - (t^2 / 2)e^2 t$
 - (b) $e^{-t} - 2e^2 t + 3te^2 t - (t^3 / 6)e^2 t$
 - (c) $-e^{-t} + 3e^2 t - t^2 e^2 t$
 - (d) $-e^{-t} + 2e^2 t - 2te^2 t + (t^3 / 3)e^2 t$
 - (e) $e^{-t} + e^2 t - 2te^2 t + (t^2 / 2)e^2 t$
4. Use Laplace's equation to find the capacitance of a parallel plate capacitor.
 - (a) $C = (\epsilon^2 S / d)$
 - (a) $C = (/ d^2)$
 - (b) $C = (/ d)$
 - (c) $C = (\epsilon / dS)$
5. In a lap winding dc machine number of conductors are 100 and number of parallel paths are 10. Find the average pitch
 - (a) 150
 - (b) 100
 - (c) 200
 - (d) 5
 - (e) 10

True / False (10 marks)

Answer True or False

6. True or False: The maximum pressure exerted on an aircraft flying at 200 mph at sea level is 19.5 psia.
7. True or False: Zinc silicate is a fluorescent material that exhibits red color fluorescence under UV light.
8. True or False: The correct answer to 'Suppose a fluid particle moves according to the equations $x = at$, $y = bt - ct^2$, where a , b , c are constants. Find the normal and...' is ' $a_n = [(2ca) / \sqrt{a^2 - (b - 2ct)^2}]$, $a_t = [\{2c(b - 2ct)\} / \sqrt{a^2 - (b - 2ct)^2}]$ '.
9. True or False: The entropy change can be expressed as $ds = C_p (dT/T) + (R/J)(dp/p)$.
10. True or False: The Laplace transform of $f(t) = \cos(kt)$ for $t > 0$ is $s / (s^2 + k^2)$, valid for $s > 0$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given the conditions of air flow at a velocity of 224 fps, a pressure of 14.7 psia, and a temperature of 17°analyzeandcalculatetheMachnumberoftheflow.Discusstheimplicationsofyourf
11. Discuss the number of entries in the truth table of a 3-input NAND gate, and analyze how the number of inputs affects the overall functionality and characteristics of the gate.

End of Paper